

ASSIGNMENT-1

QUESTION-1

Classes ,Objects and Inheritance

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QUESTIONS

Q1
Program based on Classes and Objects - Initializing Data member using Objects directly
10 marks

Q2
Classes and Objects-Initialize Object Data Using Setter Methods
10 marks

Q3
Student Details Using Default and Parameterized Constructors
10 marks

Q4
Net Salary Calculation Using Single Inheritance
10 marks

Q5
Employee Gross Salary Calculation Using Multilevel Inheritance
10 marks

5 / 5 answered

Q1 Program based on Classes and Objects - Initializaing Data member using Objects directly10 marks

Write a Java program to create a class named Student with the following data members:

```
int rollNo
String name
double marks
```

Create a method named displayDetails() to display the student's details.

In the main() method:

Create a Student object.
Read the values from the user.
Initialize the data members directly using the object (do not use a constructor or setter methods).
Invoke the displayDetails() method using the object.

Input Format

The first line contains an integer representing the roll number.
The second line contains the student's name.
The third line contains a decimal number representing the student's marks.

Output Format

Print the details in the following format:

```
Roll No: <rollNo>
Name: <name>
Marks: <marks>
```

CODE AND OUTPUT

Your Code

```
1 import java.util.Scanner;
2
3 class Student {
4
5     int rollNo;
6     String name;
7     double marks;
8
9     void displayDetails() {
10         System.out.println("Roll No: " + rollNo);
11         System.out.println("Name: " + name);
12         System.out.println("Marks: " + marks);
13     }
14
15     public static void main(String[] args) {
16
17         Scanner sc = new Scanner(System.in);
18
19         Student s = new Student();
20
21         s.rollNo = Integer.parseInt(sc.nextLine());
22         s.name = sc.nextLine();
23         s.marks = Double.parseDouble(sc.nextLine());
24
25         s.displayDetails();
26
27         sc.close();
28     }
29 }
```

✓ Submitted

Input

```
101
ravi
89.5
```

Output

```
Roll No: 101
Name: ravi
Marks: 89.5
```

Visible Test Cases

Test Case 1	✓
Test Case 2	✓
Test Case 3	✓

10.00 / 10 marks

QUESTION-2

Classes ,Objects and Inheritance

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QUESTIONS

Q1

Program based on Classes and Objects - Initializing Data member using Objects directly

10 marks

Q2

Classes and Objects-Initialize Object Data Using Setter Methods

10 marks

Q3

Student Details Using Default and Parameterized Constructors

10 marks

Q4

Net Salary Calculation Using Single Inheritance

10 marks

Q5

Employee Gross Salary Calculation Using Multilevel Inheritance

10 marks

5 / 5 answered

Q2 Classes and Objects-Initialize Object Data Using Setter Methods

10 marks

Write a Java program to create a class named Student with the following private data members:

```
int rollNo
String name
double marks
```

Create the following setter methods to initialize the data members:

```
setRollNo(int rollNo)
setName(String name)
setMarks(double marks)
```

Also, create a method named displayDetails() to display the student's details.

In the main() method:

Create a Student object.
Read the values from the user.
Initialize the data members using the setter methods only.
Call the displayDetails() method to display the student's information.

Input Format

The first line contains an integer representing the roll number.
The second line contains the student's name.
The third line contains a decimal number representing the student's marks.

Output Format

Display the details in the following format:

```
Roll No: <rollNo>
Name: <name>
Marks: <marks>
```

CODE AND OUTPUT

Your Code

```
1 import java.util.Scanner;
2
3 class Student{
4
5     private int rollNo;
6     private String name;
7     private double marks;
8
9     void setRollNo(int rollNo){
10         this.rollNo = rollNo;
11     }
12
13     void setName(String name){
14         this.name = name;
15     }
16
17     void setMarks(double marks){
18         this.marks = marks;
19     }
20
21     void displayDetails() {
22         System.out.println("Roll No: " + rollNo);
23         System.out.println("Name: " + name);
24         System.out.println("Marks: "+ marks);
25     }
26
27     public static void main(String[] args){
28         Scanner sc = new Scanner(System.in);
29
30         Student s = new Student();
31
32         s.setRollNo(sc.nextInt());
33         sc.nextLine();
34         s.setName(sc.nextLine());
35         s.setMarks(sc.nextDouble());
36
37         s.displayDetails();
38     }
39 }
```

Submitted

Input

```
101
india
95.5
```

Output

```
Roll No:101
Name:raj
Marks:95.5
```

Visible Test Cases

Test Case 1	✓
Test Case 2	✓
Test Case 3	✓

10.00 / 10 marks

QUESTION-3

QUESTIONS

Q1
Program based on Classes and Objects - Initializing Data member using Objects directly
10 marks

Q2
Classes and Objects-Initialize Object Data Using Setter Methods
10 marks

Q3
Student Details Using Default and Parameterized Constructors
10 marks

Q4
Net Salary Calculation Using Single Inheritance
10 marks

Q5
Employee Gross Salary Calculation Using Multilevel Inheritance
10 marks

5 / 5 answered

Q3 Student Details Using Default and Parameterized Constructors10 marks

Write a Java program to create a class named Student with the following data members:

```
int rollNo
String name
double marks
```

Implement the following constructors:

Default Constructor
Initializes the data members with the following values:
Roll Number: 101
Name: Rahul
Marks: 85.5

Parameterized Constructor
Accepts the roll number, name, and marks as parameters and initializes the data members.

Create a method named displayDetails() to display the student's details in the following format:

```
Roll No: <rollNo>
Name: <name>
Marks: <marks>
Task
```

In the main() method:

Create the first Student object using the default constructor and display its details.
Read the roll number, name, and marks from the user.
Create the second Student object using the parameterized constructor and display its details.

Input Format
The first line contains an integer representing the roll number.
The second line contains the student's name.
The third line contains a decimal number representing the student's marks.

Output Format
Display the details of both students.

The first student's details should be displayed using the default constructor, followed by the second student's details using the parameterized constructor.

CODE AND OUTPUT

Your Code

```
1 import java.util.Scanner;
2
3 class Student {
4
5     int rollNo;
6     String name;
7     double marks;
8
9     Student() {
10         rollNo = 101;
11         name = "Rahul";
12         marks = 85.5;
13     }
14
15     Student(int rollNo, String name, double marks) {
16         this.rollNo = rollNo;
17         this.name = name;
18         this.marks = marks;
19     }
20
21     void displayDetails() {
22         System.out.println("Roll No: " + rollNo);
23         System.out.println("Name: " + name);
24         System.out.println("Marks: " + marks);
25     }
26
27     public static void main(String[] args) {
28         Scanner sc = new Scanner(System.in);
29
30         Student s1 = new Student();
31         s1.displayDetails();
32
33         int rollNo = sc.nextInt();
34         sc.nextLine();
35         String name = sc.nextLine();
36         double marks = sc.nextDouble();
37
38         Student s2 = new Student(rollNo, name, marks);
39         s2.displayDetails();
40     }
41 }
42 }
```

Input

```
101
india
95.5
```

Output

```
Roll No: 101
Name: Rahul
Marks: 85.5
Roll No: 101
Name: india
Marks: 95.5
```

Visible Test Cases

Test Case 1	✓
Test Case 2	✓
Test Case 3	✓

10.00 / 10 marks

QUESTION-4

Classes ,Objects and Inheritance

Q1
Program based on Classes and Objects - Initializing Data member using Objects directly
10 marks

Q2
Classes and Objects-Initialize Object Data Using Setter Methods
10 marks

Q3
Student Details Using Default and Parameterized Constructors
10 marks

Q4
Net Salary Calculation Using Single Inheritance
10 marks

Q5
Employee Gross Salary Calculation Using Multilevel Inheritance
10 marks

5 / 5 answered

Q4 Net Salary Calculation Using Single Inheritance

10 marks

Write a Java program to demonstrate Single Inheritance.

Create a superclass named Employee with the following data members:

```
int empId  
String empName  
double basicSalary
```

Create a method named `getEmployeeDetails(int empId, String empName, double basicSalary)` to initialize the employee details.

Create a subclass named Salary that inherits from Employee.

In the subclass, create a method named `calculateSalary()` that calculates the following:

House Rent Allowance (HRA) = 20% of Basic Salary
Dearness Allowance (DA) = 10% of Basic Salary
Net Salary = Basic Salary + HRA + DA

Create another method named `displayDetails()` to display the employee details and the calculated salary.

Input Format

The first line contains an integer representing the Employee ID.
The second line contains the Employee Name.
The third line contains the Basic Salary.

Output Format

Display the output in the following format:

```
Employee ID: <empId>  
Employee Name: <empName>  
Basic Salary: <basicSalary>  
HRA: <hra>  
DA: <da>  
Net Salary: <netSalary>
```

CODE AND OUTPUT

Your Code

```
1 import java.util.Scanner;  
2  
3 class Employee {  
4     int empId;  
5     String empName;  
6     double basicSalary;  
7  
8     void getEmployeeDetails(int empId, String empName, double basicSalary) {  
9         this.empId = empId;  
10        this.empName = empName;  
11        this.basicSalary = basicSalary;  
12    }  
13  
14    public static void main(String[] args) {  
15        Scanner sc = new Scanner(System.in);  
16  
17        Salary s = new Salary();  
18  
19        int id = sc.nextInt();  
20        sc.nextLine();  
21        String name = sc.nextLine();  
22        double salary = sc.nextDouble();  
23  
24        s.getEmployeeDetails(id, name, salary);  
25        s.calculateSalary();  
26        s.displayDetails();  
27    }  
28 }  
29  
30 class Salary extends Employee {  
31     double hra, da, netSalary;  
32  
33     void calculateSalary() {  
34         hra = basicSalary * 0.20;  
35         da = basicSalary * 0.10;  
36         netSalary = basicSalary + hra + da;  
37     }  
38  
39     void displayDetails() {  
40         System.out.println("Employee ID: " + empId);  
41         System.out.println("Employee Name: " + empName);  
42         System.out.println("Basic Salary: " + basicSalary);  
43         System.out.println("HRA: " + hra);  
44         System.out.println("DA: " + da);  
45         System.out.println("Net Salary: " + netSalary);  
46     }  
47 }
```

Input

```
101  
indira  
50000
```

Output

```
Employee ID: 101  
Employee Name: indira  
Basic Salary: 50000.0  
HRA: 10000.0  
DA: 5000.0  
Net Salary: 65000.0
```

Visible Test Cases

Test Case 1	✓
Test Case 2	✓
Test Case 3	✓

10.00 / 10 marks

QUESTION-5

Q1
Program based on Classes and Objects - Initializing Data member using Objects directly
10 marks

Q2
Classes and Objects Initialize Object Data Using Setter Methods
10 marks

Q3
Student Details Using Default and Parameterized Constructors
10 marks

Q4
Net Salary Calculation Using Single Inheritance
10 marks

Q5
Employee Gross Salary Calculation Using Multilevel Inheritance
10 marks

5 / 5 answered

Q5 Employee Gross Salary Calculation Using Multilevel Inheritance
10 marks

Write a Java program to demonstrate Multilevel Inheritance.

Create three classes as described below:

Class 1: Employee (Superclass)

Data Members:

int empId
String empName

Method:

getEmployeeDetails(int empId, String empName) – Initializes the employee details.

Class 2: Salary (Derived Class)

This class inherits from Employee.

Data Member:

double basicSalary

Method:

getBasicSalary(double basicSalary) – Initializes the basic salary.

Class 3: GrossSalary (Derived Class)

This class inherits from Salary.

Method:

calculateSalary() – Calculates:
House Rent Allowance (HRA) = 20% of Basic Salary
Dearness Allowance (DA) = 10% of Basic Salary
Gross Salary = Basic Salary + HRA + DA

Method:

displayDetails() – Displays the employee details and the calculated salary.

Input Format

The first line contains an integer representing the Employee ID.
The second line contains the Employee Name.
The third line contains the Basic Salary.

Output Format

Display the output in the following format:

Employee ID: <empId>
Employee Name: <empName>
Basic Salary: <basicSalary>
HRA: <hra>
DA: <da>
Gross Salary: <grossSalary>

CODE AND OUTPUT

```
1 import java.util.Scanner;
2
3 class Employee {
4     int empId;
5     String empName;
6
7     void getEmployeeDetails(int empId, String empName) {
8         this.empId = empId;
9         this.empName = empName;
10    }
11
12    public static void main(String[] args) {
13        Scanner sc = new Scanner(System.in);
14
15        GrossSalary g = new GrossSalary();
16
17        int id = sc.nextInt();
18        sc.nextLine();
19        String name = sc.nextLine();
20        double basicSalary = sc.nextDouble();
21
22        g.getEmployeeDetails(id, name);
23        g.getBasicSalary(basicSalary);
24        g.calculateSalary();
25        g.displayDetails();
26    }
27 }
28
29 class Salary extends Employee {
30     double basicSalary;
31
32     void getBasicSalary(double basicSalary) {
33         this.basicSalary = basicSalary;
34     }
35 }
36
37 class GrossSalary extends Salary {
38     double hra, da, grossSalary;
39
40     void calculateSalary() {
41         hra = basicSalary * 0.20;
42         da = basicSalary * 0.10;
43         grossSalary = basicSalary + hra + da;
44     }
45
46     void displayDetails() {
47         System.out.println("Employee ID: " + empId);
48         System.out.println("Employee Name: " + empName);
49         System.out.println("Basic Salary: " + basicSalary);
50         System.out.println("HRA: " + hra);
51         System.out.println("DA: " + da);
52         System.out.println("Gross Salary: " + grossSalary);
53     }
54 }
```

102
Gayathri
50000

Output

Visible Test Cases

Test Case 1 ✓

Test Case 2 ✓

Test Case 3 ✓

10.00 / 10 marks